Paper ED-GW-00: Gravitational Waves as Propagating Saddle-Gradient Reconfigurations in the ED Substrate

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Paper_ED_GW_00 — Gravitational Waves as Propagating Saddle-Gradient Reconfigurations in the ED Substrate

Series: Wave-3 Generative Papers (Dynamics Arc — anchor paper) **Status:** Conditional structural derivation given the 13 ED primitives (Paper_087) + one paper-specific naming convention (P-GW-SaddleReconfiguration, definitional only). Verdict per Paper_095: **M3 (form-IDENTIFIED + value-INHERITED).** Row 12 (quadrupole source-amplitude formula) retroactively closed at D-via-I (2026-05-16) via the audited NoetherFlux substrate-graph chain (load-bearing #4 closure). **Date:** 2026-05-16 (revised — row 12 retroactive closure via NoetherFlux audit; see §3.9) **Anchors:** Paper_ED_SC_4_9 (substrate-action saddle Hessian classification); Paper_089 (V1 retarded kernel); Paper_090 (V5 cross-chain kernel); Paper_093 (T18 kernel-arrow); Paper_012 (P-RB-1 substrate-c); Paper_032 (WeakFieldPrereqs / 1/r dilution); Paper_033 (AcousticMetric); Paper_072 (individuation regime); Paper_109 (LorentzRepresentations); Paper_116 (MRR_MRP / massless slot); Paper_ED_CCC §3.3 (Gravitational Wave Epoch); Papers SC-4.x (scale-collapse); Paper_052_5_Pred_MergerLag. **Row 12 closure (substrate-research support):** Memo_ED_RadiationLaw_Scoping; Memo_ED_RadiationLaw_NoetherFlux; Memo_ED_RadiationLaw_NoetherFlux_Audit (audit ACCEPTED at approximately-standard-physics level); Memo_ED_LoadBearingProgram_Overview (load-bearing #4 closure context).

Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim derivation of LIGO/Virgo strain amplitudes $h \sim 10^{-21}$, observed GW frequencies, or the speed-of-GW $\approx c$ (GW170817). All three are INHERITED.
2. It does **not** claim closed-proof reduction of GR's transverse-traceless mode equations from substrate primitives. The substrate-side reading is identified; the GR-side equations are INHERITED with the correspondence at the level of structural identification.
3. It does **not** claim closed-proof reconstruction in the Hardy 2001 / CDP 2011 / Coecke-Kissinger operational-reconstruction sense. ED sits in the substrate-ontology lineage ('t Hooft cellular-automaton; Wolfram Ruliad; causal-set program), not the operational-reconstruction lineage.
4. It does **not** posit a graviton, a metric perturbation, a Hilbert-space GW mode, a gauge freedom, or a stress-energy pseudotensor. GWs in ED are propagating saddle-gradient reconfigurations carried by V1 retarded propagation; energy is local ED-gradient load. The substrate ontology replaces these GR machineries with substrate-side equivalents.
5. It does **not** introduce new substrate primitives. P-GW-SaddleReconfiguration (§2.3)

is a paper-specific *naming convention* per Paper_095 §2.3 — definitional only, no new substrate content beyond the conjunction of Paper_ED_SC_4_9 saddle Hessian content + Paper_089 V1 retarded propagation.

Abstract

Gravitational waves in ED are **propagating reconfigurations of the substrate-action saddle's Hessian-signature content**, carried by V1 retarded propagation at substrate-c. The substrate-action $S_{\text{sub}}[\Psi]$ (Paper_ED_SC_4_9) admits saddles whose Hessian eigenvalue-signature partitions each substrate locus into compression-dominant and expansion-dominant orthogonal axes; a GW is the coherent transverse oscillation in which these roles swap periodically as the V1-propagated wavefront passes. Two transverse polarizations follow from the two independent Hessian-flip directions at a generic saddle in $D = 3 + 1$ (Paper_ED_SC_4_9 + Paper_116 + Paper_109); no scalar / vector modes admitted. Propagation at substrate-c (Paper_012 + Paper_089) is forward-causal (Paper_093 T18); 1/r amplitude dilution inherits from Paper_032; no dispersion at cosmic scales from SC-4.x + Paper_ED_CCC §3.6. **No graviton, no metric perturbation, no Hilbert-mode, no gauge freedom, no pseudotensor.** Verdict per Paper_095: **M3 (form-IDENTIFIED + value-INHERITED). F1 (sharpest):** detection of non-transverse polarization modes (scalar / vector) beyond the saddle's two Hessian-flip directions — refutes the substrate composition.

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§1 Statement of Result

A gravitational wave at substrate level is the structural composition of three elements:

1. **Substrate-action saddle Hessian content** (Paper_ED_SC_4_9): $S_{\text{sub}}[\Psi]$ admits saddles whose Hessian partitions each locus into compression-dominant and expansion-dominant orthogonal axes.

2. **V1 retarded propagation** (Paper_089): finite-width retarded kernel carrying substrate content forward-causally at substrate-c (Paper_012, Paper_093 T18).

3. **Coherent transverse reconfiguration**: the saddle's Hessian eigenvalue-signature swaps role periodically between orthogonal transverse axes as the V1-propagated wavefront advances. Two transverse polarizations correspond to the two independent Hessian-flip directions.

"The saddle is the geometry; the gradients are the calculus; the boundaries are the topology; the horizons are the dynamics." A GW is a **dynamics-class** phenomenon — propagating horizons / saddles in motion — not a kinematics-class perturbation on a fixed background. The substrate has no fundamental metric to perturb; what propagates is the saddle-Hessian-signature pattern itself.

Inherited observational consequences: speed = c (Paper_012 + Paper_089); two transverse polarizations only (Paper_ED_SC_4_9 + Paper_116 + Paper_109); $h \sim 1/r$ amplitude (Paper_032); no cosmic-scale dispersion (SC-4.x + Paper_ED_CCC §3.6). Empirical anchors: LIGO/Virgo/KAGRA strain spectra; GW170817 speed equality with c.

Verdict: M3 (form-IDENTIFIED + value-INHERITED) per §4 audit table. Zero pure-D rows; **seven substantive D-via-I composition rows** (row 12 newly closed via NoetherFlux audit, 2026-05-16); one definitional naming convention; **one remaining OPEN**

item (row 13 — full inspiral-merger-ringdown waveform; Dynamics-Arc / Black-Hole-Arc follow-up; not load-bearing for M3 framing). **No remaining load-bearing OPEN items.**

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§2 Primitive Inputs + Upstream Dependencies

Primitives (Paper_087): P02, P04, P07, P09, P11, P12, P13.

Upstream: Paper_012 (substrate-c P-RB-1); Paper_032 (weak-field 1/r dilution); Paper_033 (acoustic-metric analogue-gravity); Paper_072 (individuation regime); Paper_089 (V1 retarded + T18 forward-causal); Paper_090 (V5 cross-chain); Paper_093 (T18 kernel-arrow); Paper_ED_SC_4_9 (substrate-action S_{sub}, Hessian classification); Paper_096 (cross-scale invariance + $\xi_{canonical}$); Paper_098_5 / T1 ($D = 3 + 1$); Paper_109 (Lorentz reps / helicity); Paper_116 (MRR_MRP massless slot — two-mode structure); Papers SC-4.x (scale-collapse, cosmic homogeneity); Paper_ED_CCC §3.3 (cosmic-scale GW Epoch); Paper_ED_CCC §3.6 (post-SCBU homogeneity); Paper_052_5_Pred_MergerLag (PROVISIONAL empirical-anchor reference); LIGO/Virgo/KAGRA observational data (INHERITED).

Paper-specific naming convention (no new postulate):

- **P-GW-SaddleReconfiguration:** Naming convention for the conjunction *"GW = propagating reconfiguration of substrate-action saddle Hessian content, carried by V1 retarded propagation."* Definitional per Paper_095 §2.3; no new substrate content beyond the Paper_ED_SC_4_9 + Paper_089 conjunction.

No new substrate primitives. No paper-specific postulates beyond the naming convention.

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§3 Derivation

3.1 Saddles as ED geometry

Per Paper_ED_SC_4_9, $S_{\text{sub}}[\Psi]$ admits critical points whose Hessian $\mathcal{H} = \delta^2 S_{\text{sub}}/\delta\Psi\delta\Psi'$ has mixed eigenvalue signature at saddle loci. Each saddle locus partitions into **compression-dominant** axes (negative Hessian eigenvalues, substrate-action concavity) and **expansion-dominant** axes (positive Hessian eigenvalues, substrate-action convexity). Boundaries between regions of differing Hessian signature are substrate boundaries; surfaces where Hessian-flipped directions meet are substrate horizons. Per the corpus's compact statement, *the saddle is the geometry; the gradients are the calculus; the boundaries are the topology; the horizons are the dynamics.* GW phenomenology lives in the **dynamics class** — propagating saddles / horizons in motion — not kinematics on a fixed background.

3.2 What "a wave" means in ED

ED has no fundamental metric; the acoustic metric of Paper_033 is a derived analogue-gravity object on substrate, not a fundamental field. The standard GR reading "wave = metric perturbation $h_{\mu\nu}$ on background $g_{\mu\nu}$" is not available substrate-side. The substrate-side reading: **a wave is a V1-propagated reconfiguration of substrate-action content**, with no background/perturbation split. The substrate-action saddle structure itself advances through substrate loci as the wave; there is no "fixed geometry plus small ripple" decomposition.

3.3 V1 propagation of saddle-Hessian reconfiguration

A GW is a substrate-action saddle whose Hessian-signature content is non-stationary — the eigenvalue-pattern at a given locus changes coherently as a function of substrate-time. Paper_089's V1 retarded kernel carries this changing Hessian content forward at substrate-c (Paper_012 P-RB-1). Forward causality is supplied by Paper_093 T18 (kernel-arrow direction); advanced V1 is refuted by Paper_089 T18 + P11. The transverse character follows from saddle geometry: the propagation direction is the "carrier" structure, while the Hessian-signature partition lies in the orthogonal subspace where reconfiguration occurs.

This is the substrate-side content of P-GW-SaddleReconfiguration (§2.3): the Paper_ED_SC_4_9 + Paper_089 conjunction named definitionally.

3.4 Compression–expansion oscillation

At each locus through which a GW propagates, the saddle Hessian's compression-dominant and expansion-dominant orthogonal axes alternate dominance over the wave period: axis A compression-dominant at time t_1 becomes expansion-dominant at $t_1 + T/2$, with axis B swapping in inverse phase. After a full period T, the original configuration returns. LIGO's observed transverse stretch/squeeze pattern at orthogonal arms is the substrate-side observable correlate; see §3.8 table for the GR vs ED comparison.

3.5 Energy transport as ED-gradient redistribution

GR's stress-energy pseudotensor for GW energy is gauge-dependent (Misner-Thorne-Wheeler §20). The substrate-side reading: **GW energy is local ED-gradient load** — the substrate-action content concentrated at the locus where the saddle-Hessian reconfiguration is currently propagating. This is a substrate-graph quantity, not a tensorial object; it requires no gauge fixing. The substrate-graph derivation of the quantitative source-amplitude formula (substrate-side analog of GR's quadrupole formula, with $\sim h^2\omega^2$ scaling) is identifiable from Paper_032 + V1 propagation + saddle-source composition; **OPEN** at quantitative-derivation level (audit row 12).

3.6 Amplitude falloff

A GW source at substrate-graph locus ℓ_0 emits saddle-reconfiguration content carried outward by V1 retarded propagation. Paper_032 weak-field prerequisites + substrate-side spherical-source dilution $\rightarrow$ amplitude $h(r) \sim 1/r$ at substrate-graph distance r. Numerical content (source amplitude, distance, observed strain) is INHERITED. Substrate-side and GR-side give identical $h \sim 1/r$ predictions; this is the standard inverse-distance dilution and matches LIGO sensitivity scaling.

3.7 Polarization structure

A saddle in $D = 3+1$ (Paper_098_5 / T1) has, at each locus, a Hessian with eigenvalue-signature partitioning 3 spatial + 1 time-like directions. Of the 3 spatial directions: one coincides with the V1 propagation direction (the kernel-arrow, the "carrier" structure); the remaining 2 are transverse to propagation. The two transverse directions admit independent Hessian-eigenvalue-flip patterns, giving the **"+" polarization** (orthogonal axes alternate Hessian-flip in phase) and the **"×" polarization** (same pattern rotated 45°). No third polarization is structurally admitted in $D = 3 + 1$; the longitudinal direction does not admit independent Hessian-flip content (it is the wave's carrier, not a reconfigurable axis), excluding scalar / longitudinal modes at substrate level. Substrate composition: Paper_ED_SC_4_9 saddle Hessian + Paper_116 massless-slot two-mode structure + Paper_109 Lorentz reps + $D = 3+1$ (T1) + V1 propagation direction. Same polarization count as GR's transverse-traceless modes $(h_+, h_\times)$; same exclusion of scalar/vector modes.

3.8 Comparison with GR

Element	GR reading	ED substrate reading	Status
What propagates	Metric perturbation $h_{\mu\nu}$ on background $g_{\mu\nu}$	Saddle-Hessian-signature reconfiguration carried by V1 retarded kernel	**Replaced** (§3.2, §3.3)
Propagation speed	c (Einstein field equations)	Substrate-c (Paper_012 + Paper_089)	**Inherited** (GW170817 confirms)
Polarization count	2 (TT modes h_+, $h_\times$)	2 (two transverse Hessian-flip directions, §3.7)	**Inherited**
Amplitude falloff	$h \sim 1/r$ from linearized GR	$h \sim 1/r$ from Paper_032 + V1 spherical dilution	**Inherited**
Energy content	Pseudotensor (gauge-dependent)	Local ED-gradient load	**Replaced** (§3.5)
Source mechanism	Stress-energy quadrupole formula	Substrate-action saddle motion via M3-template applied to time-varying Ψ sources; quadrupole formula closes via NoetherFlux substrate-graph chain (§3.9)	**Replaced** + **D-via-I (row 12 closed)**
Particle content	Graviton (spin-2 massless boson, hypothetical)	None at substrate level	**Replaced**
Gauge freedom	Diffeomorphism + linearized gauge $(h_{\mu\nu} \to h_{\mu\nu} + \partial_{(\mu}\xi_{\nu)})$	None at substrate level	**Replaced**
Cosmic dispersion	None at standard GR linear order	None at substrate level (SC-4.x + Paper_ED_CCC §3.6)	**Inherited**

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§3.9 Quadrupole Source-Amplitude Closure (Row 12 — Added 2026-05-16)

Audit row 12 (substrate-graph derivation of the source-amplitude formula, substrate analog of the GR quadrupole) closes retroactively via the audited NoetherFlux substrate-graph chain — load-bearing #4 of the cosmology/dynamics program (Memo_ED_LoadBearingProgram_Overview). The closure is at D-via-I via the M3-template extended to time-varying Ψ sources.

The substrate-graph chain (Memo_ED_RadiationLaw_NoetherFlux; audit ACCEPTED at approximately-standard-physics level per Memo_ED_RadiationLaw_NoetherFlux_Audit):

Step	Substrate-side content
A	Time-varying mass-quadrupole source: substrate-side $Q_{ij}^{\mathrm{sub}}(\ell,t) = \int(\Psi \text{ density})(x_i x_j - \frac{1}{3}\delta_{ij}\lvert x\rvert^2)d^3x$ with non-trivial $\dddot{Q}_{ij}$
B	Substrate-side Noether stress-energy $T^{\mu\nu}_{\mathrm{sub}}$ via P03 + P13 substrate-translation invariance (Paper_087) + Paper_ED_SC_4_9 S_{sub} functional. For time-varying sources, flux components T^{0i}_{sub} scale with $\lvert\dddot{Q}_{ij}\rvert^2$ at appropriate retardation
C	DCGT (Paper_073) coarse-grains substrate-side flux to continuum stress-energy flux T^{0i}_{eff} within hydrodynamic-window scale-separation
D	Linearized Einstein retarded-Green's-function machinery inherits (standard GR)
E	$P_{\mathrm{GW}} = \frac{G}{5c^5}\langle\dddot{Q}_{ij}\dddot{Q}^{ij}\rangle$ — standard quadrupole formula with G via Paper_027 (substrate-side dimensional rearrangement) + c via Paper_012

Closure qualifications (per audit): substrate-side source-class identification (time-varying multipole) and stress-energy flux form (quadrupole-flux) are INHERITED by standard-GR-analog applied to substrate-side $\mathcal{L}_{\mathrm{sub}}$; DCGT applies cleanly for radiation at standard physical frequencies; edge-of-validity at near-Planck-scale frequencies. Closure level matches load-bearing #3 audit acceptance pattern.

Substrate-c bound preserved at substrate-graph level throughout the chain. Quadrupole formula is continuum-side hydrodynamic flux through DCGT — same template as the M3 chain that closed load-bearing #1 (inflation arc) and load-bearing #3 (horizon-motion law).

The GW paper's only substantive load-bearing OPEN closes. Remaining row 13 (substrate-graph derivation of full inspiral-merger-ringdown waveform from BH dynamics) is a Dynamics-Arc / Black-Hole-Arc follow-up, not load-bearing for the M3 framing.

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§4 Audit Table

#	Step	Label	Notes
1	P02, P04, P07, P09, P11, P12, P13	P	Paper_087.

#	Step	Label	Notes
2	Upstream inheritance block	I	Paper_012 (substrate-c); Paper_032 (1/r dilution); Paper_033 (acoustic metric); Paper_072 (individuation); Paper_089 (V1 retarded + T18); Paper_090 (V5); Paper_093 (T18 kernel-arrow); Paper_ED_SC_4_9 (substrate-action saddle Hessian); Paper_096 ($\xi_{canonical}$); Paper_098_5 / T1 ($D = 3 + 1$); Paper_109 (Lorentz reps); Paper_116 (massless-slot two-mode); Papers SC-4.x + Paper_ED_CCC §3.6 (scale-collapse + cosmic homogeneity); Paper_ED_CCC §3.3 (GW Epoch).
3	**P-GW-SaddleReconfiguration** (naming convention)	**P**	§2.3; definitional per Paper_095 §2.3; no new substrate content.
4	GW = propagating saddle-Hessian reconfiguration	**D-via-I**	Composition of Paper_ED_SC_4_9 + Paper_089 under P-GW-SaddleReconfiguration; §3.3.
5	Coherent transverse compression/expansion role-swap at orthogonal axes	**D-via-I**	Composition of saddle Hessian eigenvalue-signature partition + propagating reconfiguration; §3.4.

#	Step	Label	Notes
6	Two transverse polarization modes; no scalar / vector / longitudinal	**D-via-I**	Composition of Paper_ED_SC_4_9 + Paper_116 + Paper_109 + T1 + V1 propagation direction; §3.7.
7	Propagation speed = substrate-c	**I**	Direct inheritance: substrate-c is the V1 propagation rate by Paper_012 + Paper_089. §3.3, §3.8.
8	Amplitude $h \sim 1/r$ at weak field / large distance	**D-via-I**	Composition of Paper_032 + V1 spherical dilution; §3.6.
9	No dispersion at cosmic scales	**D-via-I**	Composition of SC-4.x + Paper_ED_CCC §3.6 + V1; §3.8.
10	Energy = local ED-gradient load (no pseudotensor)	**D-via-I**	Composition of saddle Hessian content + V1 propagation; §3.5.
11	No graviton; no metric perturbation; no Hilbert-mode; no gauge freedom	**A→position**	Four substrate-ontology replacement claims bundled per Paper_095 §3.3 framing-claim verdict labels. §3.8, §6.
12	Substrate-graph derivation of source-amplitude formula (substrate analog of GR quadrupole)	**D-via-I (CLOSED 2026-05-16)**	§3.9. Closes via NoetherFlux substrate-graph chain (load-bearing #4 closure); audit ACCEPTED at approximately-standard-physics level per Memo_ED_RadiationLaw_NoetherF Same M3-template pattern as loads #1, #3.
13	Substrate-graph derivation of full inspiral-merger-ringdown waveform from BH dynamics	**OPEN**	Dynamics-Arc / Black-Hole-Arc follow-up; **not load-bearing for M3 framing.**

#	Step	Label	Notes
14	Verdict M3 (form-IDENTIFIED + value-INHERITED)	**A→position**	Per Paper_095 §3.3 line 77; framing-claim verdict row.

Zero pure-D rows. **Seven substantive D-via-I rows** (4, 5, 6, 8, 9, 10, 12 — row 12 newly closed via NoetherFlux audit). **One remaining OPEN item** (row 13 — full inspiral-merger-ringdown waveform; not load-bearing for M3 framing; Dynamics-Arc / Black-Hole-Arc follow-up). One naming convention (3); no new postulates or primitives. **The GW paper has no remaining load-bearing OPEN items.**

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§5 Falsification Criteria

- **F1 (sharpest):** Detection of non-transverse GW polarization modes — scalar (breathing / longitudinal) or vector — beyond the two transverse Hessian-flip directions admitted by saddle geometry. Refuted either empirically (LIGO/Virgo polarization-content analyses identifying a non-transverse mode) **or** via substrate-graph derivation showing the saddle Hessian admits a third independent transverse mode in $D = 3+1$. Refutes §3.7. (Currently consistent: LIGO/Virgo polarization analyses constrain non-tensor modes.)
- **F2:** Observation of cosmic-scale GW dispersion — a frequency-dependent propagation speed beyond standard-GR linear order — that cannot be absorbed into substrate-c content. Refutes §3.8 no-dispersion claim. (Currently consistent: GW170817 multi-messenger bound $|c_{\mathrm{GW}} - c|/c < 10^{-15}$.)
- **F3:** Observed GW amplitude falloff inconsistent with $h \sim 1/r$ scaling at large distance — e.g., $h \sim 1/r^{1+\epsilon}$ for non-trivial ϵ — that cannot be absorbed into source-modeling uncertainty. Refutes §3.6.
- **F4:** Detection of a graviton-particle signature in any quantum-gravity experiment — a substrate-side mode behaving as an irreducible spin-2 particle quantum and not as a substrate-action saddle reconfiguration. Refutes the no-graviton replacement of §3.8 and the substrate-ontology framing.

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§6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton; Wolfram Ruliad; causal-set program), **not** within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

Gravitational waves in ED are not metric perturbations on a fixed background — substrate-side there is no fundamental metric to perturb. They are propagating reconfigurations of the substrate-action saddle's Hessian-signature content, carried by V1 retarded propagation at substrate-c. *The saddle is the geometry; the gradients are the calculus; the boundaries are the topology; the horizons are the dynamics.* GWs are dynamics-class — propagating horizons / saddles in motion. The substrate-ontology replacement of GR's machinery (metric perturbation → saddle-Hessian reconfiguration; pseudotensor → ED-gradient load; graviton → no particle; gauge freedom → no gauge content; quadrupole source → substrate-action saddle motion) is structurally clean and consistent with the modern ED architecture.

Observational content (speed, polarization count, amplitude scaling, frequency, strain, no dispersion) is **INHERITED** and reproduces LIGO/Virgo/KAGRA observations and GR predictions at the structural level. Ontological machinery is **REPLACED** by substrate-side equivalents; same predictions, different ontological language, substantially leaner content (no graviton, no gauge, no pseudotensor).

Verdict: M3 (form-IDENTIFIED + value-INHERITED) per Paper_095. Seven substantive D-via-I rows in the audit table (row 12 newly closed via NoetherFlux audit, 2026-05-16); zero pure-D rows; one definitional naming convention (P-GW-SaddleReconfiguration) with no new substrate content; **one remaining OPEN item** (row 13 — substrate-graph derivation of full inspiral-merger-ringdown waveform from BH dynamics; Dynamics-Arc / Black-Hole-Arc follow-up; not load-bearing for the M3 framing). **The GW paper has no remaining load-bearing OPEN items.** Row 12 closure via NoetherFlux substrate-graph chain (§3.9; load-bearing #4 of the cosmology/dynamics program) supplies the substrate-graph derivation of the quadrupole source-amplitude formula via M3-template extended to time-varying Ψ sources — same pattern as load #1 (inflation, Paper_ED_Cos_01 M3 upgrade) and load #3 (horizon motion).

End Paper_ED_GW_00.